

# MD NAZRUL ISLAM

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## Research Focus:

Vision-language models: spatial and geometric reasoning. Multimodal benchmarks: whether a score measures the model or the metric. Low-resource NLP: what language models depend on when input is noisy and non-Latin-script. Across all three: locating a model's failure before choosing the fix.

## Education:

**Khulna University of Engineering & Technology (KUET), Khulna, Bangladesh**

January 2018 – February 2023

- B.Sc. in Computer Science & Engineering
- CGPA: 3.68 out of 4.00 (ranked in the top **20%** of the Department of Computer Science and Engineering)

## Publications:

### Under Review / In Preparation:

1. Md Asiful Islam Miah\*, **Md Nazrul Islam**\*, S M Taslim Uddin Raju, Hamdi Altaheri. "[GeoToken: Differentiable 3D Voxel Tokenization for Geometric Spatial Reasoning in VLMs](#)." Submitted to AAAI 2027. \**Equal contribution*.
  - **Contribution:** Built the training pipeline: Kaggle compute, versioned checkpointing via Hugging Face Hub. Implemented the spatially aware projection encoder that produces voxel-centroid tokens for the 4-bit LoRA-tuned Gemma decoder.
2. **Md Nazrul Islam**, Md Habibullah Belali. "[Low Scores Without a Perceptual Deficit: Separating Answer-Position Bias from Perception on MMVP](#)." Submitted to ICCIT 2026.
  - **Contribution:** Problem formulation, the reversed-options control, statistical design, analysis and writing.
3. **Md Nazrul Islam**. "[Anchored to the Form: Bangla Named Entity Recognition Depends on Entity Orthography, Not Context](#)." In preparation for *ACL Rolling Review* (October 2026), targeting NAACL 2027.
  - **Contribution:** Problem formulation, controlled perturbation design, experiments, analysis and writing.

### Journals:

1. Ovishake Sen, Mohtasim Fuad, **Md Nazrul Islam**, Jakaria Rabbi, Mehedi Masud, Md Kamrul Hasan, Md Abdul Awal, Awal Ahmed Fime, Md Tahmid Hasan Fuad, Delowar Sikder, Md Akil Raihan Iftee. "[Bangla Natural Language Processing: A Comprehensive Analysis of Classical, Machine Learning, and Deep Learning-Based Methods](#)." *IEEE Access* 10 (2022): 38999–39044.
  - **Contribution:** Co-designed the eleven-task taxonomy; wrote the deep-learning method comparisons and the low-resource challenges analysis.
2. Md Tahmid Hasan Fuad, Awal Ahmed Fime, Delowar Sikder, Md Akil Raihan Iftee, Jakaria Rabbi, Mabrook S Al-Rakhami, Abdu Gumaie, Ovishake Sen, Mohtasim Fuad, **Md Nazrul Islam**. "[Recent Advances in Deep Learning Techniques for Face Recognition](#)." *IEEE Access* 9 (2021): 99112–99142.
  - **Contribution:** Analysis of representation learning methods, loss functions, and datasets.

### Conferences:

1. Towsif Ahamed, **Md Nazrul Islam**, S M Taslim Uddin Raju, M M A Hashem. "[Blood Donor Arrival Forecasting Using Regression Model and Analysis of Donor Behavioural Pattern](#)." In *2022 25th International Conference on Computer and Information Technology (ICCIT)*, pp. 897–902. IEEE, 2022.
  - **Contribution:** Reframed a 2.67%-base-rate classification task as aggregate regression over propensity buckets; built the decision-tree forecaster and the donor-behavior analysis; evaluated the OneBlood deployment.

## Research Experience:

- **GeoToken: 3D Voxel Tokenization for Spatial Reasoning in VLMs** | PyTorch, Gemma 2026
  - Standard VLM tokenizers flatten 3D structure into 2D patches and discard the geometric relations spatial reasoning depends on; GeoToken tokenizes voxel space directly and keeps that structure intact through to the language model.
  - Two-stage architecture: a spatially aware projection encoder producing metric voxel-centroid tokens, consumed by a 4-bit LoRA fine-tuned Gemma decoder; 81.33% on SpatialRGPT-Bench at 1B parameters in 3.6 GB.
- **Low Scores Without a Perceptual Deficit: Position Bias on MMVP** | Independent research 2026
  - MMVP's below-chance scores are read as a perceptual deficit in small VLMs. The scores track answer-position bias in the benchmark's construction: for LLaVA-1.5, 93% of failures pick the same answer slot whatever the image shows.
  - A reversed-options control, one extra forward pass per question, separates slot bias from perceptual accuracy; perceptual accuracy is 0.68–0.73 across models whose published scores span 6.0 to 33.3.
- **Anchored to the Form: Bangla NER Depends on Orthography, Not Context** | Independent research 2025 – 2026
  - Bangla NER models are assumed to generalize from context the way English models do. Under controlled character perturbation they depend on entity orthography instead, 9.3 to 1 over context, and the dependence breaks under exactly the noise low-resource deployment produces.

- Quantified the context-versus-orthography gap across perturbation types and showed the diagnosis predicts which defense works: character augmentation recovers up to 14.3 F1, gradient-based adversarial training recovers nothing distinguishable from seed noise.
- [Adversarial Robustness in Low-Resource NLP \(Bangla\)](#) | PyTorch 2025
  - Character-, word-, and noise-level attacks on Bangla language models, measuring which perturbations break which encoders; the groundwork for the entity-orthography study above.
- [Multiclass SQL Injection Detection with a 1D CNN–TinyBERT Model](#) | PyTorch 2024 – 2025
  - A 1D CNN–TinyBERT classifier for multiclass SQL-injection payloads, evaluated with macro-F1 and PR-AUC because the payload classes are heavily imbalanced and accuracy would hide the minority-class failures.
- [Lightweight LLM and Reinforcement Learning for Text-to-SQL Generation](#) | PyTorch 2023
  - A TinyBERT text-to-SQL system on Spider; reinforcement learning from execution feedback raised logical correctness without increasing model size.
- **Undergraduate Thesis: BCI-Oriented EEG Signal Analysis and Classification** | TensorFlow 2022 – 2023
  - ERP detection on BCI Competition data, where the bottleneck was cross-subject generalization rather than single-subject accuracy.
  - Compared frequency and time-frequency features (PSD, WPD, STFT) after FIR, notch and CAR preprocessing for their effect on the cross-subject gap; NCA and Relief-F selection cut the feature set before classification.
  - The final CNN reached 92% cross-subject ERP accuracy; stage-by-stage removal attributed most of the gain to preprocessing and selection rather than the classifier.

## ***Professional Experience:***

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**Lecturer, Department of CSE** February 2024 – Present

- Southeast University, Dhaka, Bangladesh
  - Teaching undergraduate courses in programming, numerical methods, and introductory artificial intelligence.
  - Supervising undergraduate projects aligned with machine learning and NLP research.

**Lecturer, Department of CSE** July 2023 – February 2024

- Uttara University, Dhaka, Bangladesh

**Part-time Lecturer, Department of CSE** March 2023 – July 2023

- Central University of Science and Technology, Dhaka, Bangladesh

## ***Standardized Test Scores:***

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**IELTS Academic, 11 October 2024 – Overall Band Score: 7.5**

- Listening 7.5, Reading 8.0, Writing 6.5, Speaking 7.0 | Reference A3-BD001-A-12054929, TRF 24BD507339ISLM001A

**GRE General Test: scheduled; score to follow**

## ***Technical Skills:***

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- *Programming Languages:* C++, Java, Python
- *Machine Learning and Deep Learning Toolkit:* PyTorch, TensorFlow, NumPy, SciPy, scikit-learn, Pandas, OpenCV
- *Version Control and Documentation:* Git, LaTeX, Overleaf

## ***Honors and Achievements:***

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- Ranked **10<sup>th</sup>** globally and **1<sup>st</sup>** among Bangladeshi teams in the *Indian Rover Design Challenge (IRDC) 2020*, organized by *Mars Society South Asia (MSSA)*, as a member of **Team Durbar**, KUET.
- **Dean's Award**, Faculty of Electrical and Electronic Engineering, KUET, academic years **2018–2019** and **2019–2020**.

## ***Leadership and Volunteer Experience:***

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- **Vice President**, Hardware Acceleration Club of KUET (HACK) March 2022 – March 2023
  - Ran the club's microcontroller workshops and organized its events.
- **Programmer, Software Sub-Team**, KUET Mars Rover (Team Durbar) March 2019 – December 2022
  - Senior Programmer: rover navigation and computer vision, including a lightweight CNN for object recognition.
  - Junior Programmer: worked with Arduino and Raspberry Pi to implement key rover systems.

## ***References:***

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Available upon request.